Natural Language Processing
Assignment 1

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1 Dataset & Setup

To classify news articles into one of four categories (world, sports, business, and sci/tech), we use the AG News dataset from Hugging Face [Zhang et al.(2015)Zhang, Zhao, and LeCun]. This dataset consists of news articles with associated labels, titles, and descriptions. Each class contains 30,000 training samples and 1,900 testing samples, resulting in a total of 120,000 training samples and 7,600 testing samples.

We used the official Hugging Face split (120,000 train and 7,600 test). The training set was split into a new training set (of size 108,000) and a development set (of size 12,000) using stratified sampling with a fixed random seed of 67 to ensure reproducibility. The final split sizes are: train 108,000, dev 12,000, and test 7,600.

Preprocessing. We combined the title and description of each news article into a single text input, which was then preprocessed. First we lowercased the text, removed extra whitespace, and removed punctuation. We also removed English stopwords using the NLTK library, as they often carry less information and can introduce noise into the model. Finally, we applied lemmatization using NLTK's WordNetLemmatizer to reduce words to their dictionary forms, which helps to reduce the dimensionality of the feature space and improve model performance.

Feature extraction. Tokenization is implicitly handled by the TF-IDF vectorizer, which converts the preprocessed text into a matrix of TF-IDF features. We configured the vectorizer to consider both unigrams and bigrams (n-gram range of (1, 2)) to capture more contextual information than unigrams alone while still being less sparse than bigrams alone. We also set a maximum feature limit of 10,000 and a minimum document frequency of 2 to further reduce noise and focus on more informative features. The TF-IDF features were computed based on the training set vocabulary to avoid data leakage, and then applied to the development and test sets using the same fitted vectorizer.

2 Models

We implemented two models for the text classification task: Logistic Regression and Linear Support Vector Machine (SVM), both using Term Frequency-Inverse Document Frequency (TF-IDF) for feature extraction. The architecture of both models is similar, with the main difference being the linear classifier used for classification. The input text is first preprocessed and then transformed into TF-IDF features, which are then fed into either a Logistic Regression classifier or a Linear SVM for classification (see Figure 1 for a visual representation of the architecture).

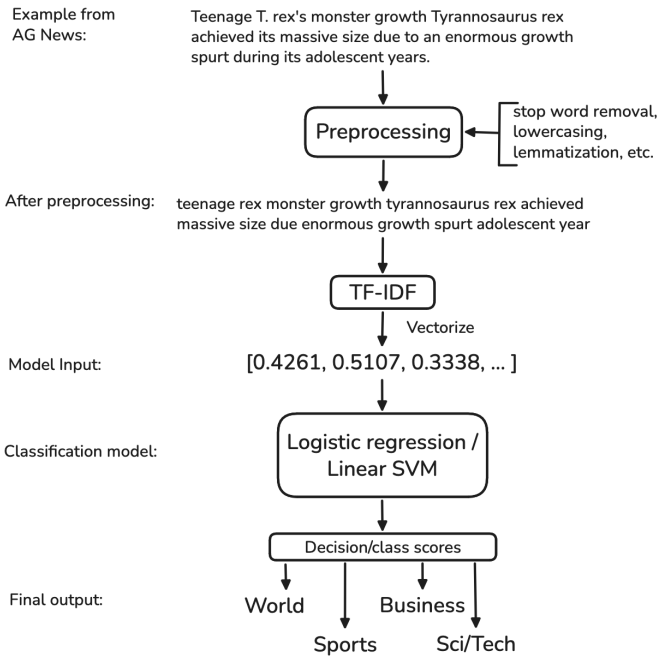

Figure 1: Model architecture for TF-IDF + Logistic Regression and TF-IDF + LinearSVM. The figure shows a random sample from the training set, which is preprocessed and transformed into TF-IDF features before being fed into the classifiers.

3 Experimental Protocol

We first built the vocabulary for the TF-IDF vectorizer using only the training split to avoid data leakage. Then, we trained each model on the training split and evaluated their performance on the development split to select the best hyperparameters based on accuracy. Finally, we performed a final evaluation on the test split using the best hyperparameters selected from the development set to report our final results.

Hyperparameter Search. We performed hyperparameter tuning for both models, selecting the best parameters based on the highest accuracy on the development set. For Logistic Regression, we searched over the regularization strength parameter C with values in $\{0.01, 0.1, 1.0, 10.0, 100.0\}$. For Linear SVM, we searched over both the regularization strength parameter C with values in $\{0.01, 0.1, 1.0, 10.0, 100.0\}$ and the loss function with options 'hinge' and 'squared_hinge'. Both models were trained with a maximum of 1,000 iterations to ensure convergence.

We recorded the accuracy, macro-F1 score, and confusion matrix for both models on the test set for the final evaluation. Both models were trained and evaluated using the same train/dev/test splits, the same preprocessing and TF-IDF feature settings, and the same random seed (67) to ensure a fair comparison between the models.

4 Results

We hyperparameter tuned both models on the dev set, selecting the best parameters based on the highest accuracy. For the Linear SVM, the best parameters were C=1.0 and loss='hinge', while for Logistic Regression, the best parameter was C=1.0. We then evaluated both models on the test set using these best parameters and report the results below.

In Table 1, we can see that the Linear SVM model achieved an overall accuracy of 0.92 and a macro-F1 score of 0.92 on the test set. The Logistic Regression model, as shown in the subsequent table (Table 2), achieved an overall accuracy of 0.91 and a macro-F1 score of 0.91 on the test set.

	Precision	**Recall**	**F1-Score**	**Support**
World	0.94	0.90	0.92	1900
Sports	**0.95**	**0.98**	**0.97**	1900
Business	0.89	0.88	0.89	1900
Sci/Tech	0.89	0.90	0.90	1900
Accuracy			0.92	7600
Macro avg	0.92	0.92	0.92	7600
Weighted avg	0.92	0.92	0.92	7600

Table 1: Classification report for Linear SVM on test set.

In the confusion matrices (Figures 2 and 3), we can observe that both models have similar patterns of misclassification, with the most common confusion occurring between the sci/tech and business classes. This is reflected in the classification reports where the precision, recall, and F1-scores for these classes are lower compared to the sport and world classes. Overall, the Linear SVM model performs slightly better than the Logistic Regression model on both accuracy and macro-F1, but the difference is not very large (e.g., 0.01 in accuracy and macro-F1).

	Precision	Recall	F1-Score	Support
World	0.93	0.91	0.92	1900
Sports	**0.95**	**0.98**	**0.96**	1900
Business	0.88	0.89	0.88	1900
Sci/Tech	0.89	0.89	0.89	1900
Accuracy			0.91	7600
Macro avg	0.91	0.91	0.91	7600
Weighted avg	0.91	0.91	0.91	7600

Table 2: Classification report for Logistic Regression on test set.

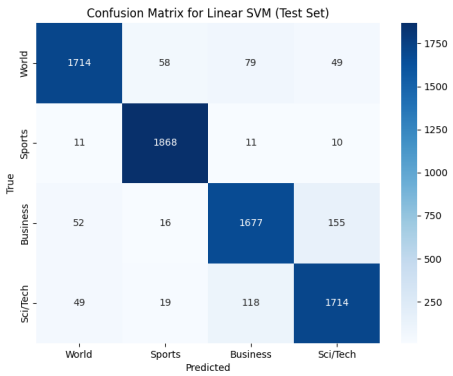

Figure 2: Confusion Matrix for Linear SVM on Test Set

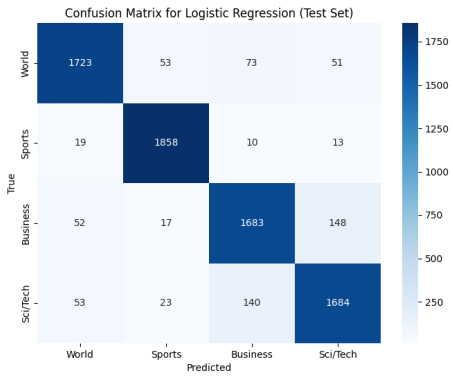

Figure 3: Confusion Matrix for Logistic Regression on Test Set

Comparing the two models in Table 3, we can see that the Linear SVM model has a slightly higher accuracy and macro-F1 score compared to the Logistic Regression model on the test set. However, both models perform quite well, with accuracies and macro-F1 scores above 0.90, indicating that the TF-IDF features are effective for this classification task.

Model	Accuracy	Macro-F1
Linear SVM	**0.92**	**0.92**
Logistic Regression	0.91	0.91

Table 3: Comparison of Accuracy and Macro-F1 for Linear SVM and Logistic Regression on Test Set

5 Error Analysis

Looking at Figures 4 and 5, we can see that the TF-IDF features form many distinct clusters in the t-SNE projection, the classes seem to be mostly separable, but there is some overlap in the center of the projection. The confusion matrices show that the most common misclassification is between the sci/tech and business classes, which is reflected in the t-SNE projection where the clusters for these classes overlap more than the others. This suggests that there are some samples in the test set that have features that are similar to both classes, which might explain why they are misclassified.

To further analyze this, we recorded 20 misclassified samples for each of the two most common confusions (i.e., business predicted as sci/tech, and sci/tech predicted as business) and looked at the raw text to see if there are any common features that might explain the confusion. We found that the errors can be put into 5 categories: 1) labeling errors, 2) ambiguous samples, 3) samples with overlapping keywords, 4) samples with insufficient context, and 5) samples that are simply difficult to classify.

Labeling errors account for a small but notable portion of the misclassified samples. These are articles that appear to have been assigned the wrong category in the AG News dataset itself, independent of any model behavior. For example, the article "Teenage T. rex's monster growth" describes paleontological research and is clearly a science article, yet it carries a Business label in the dataset. Similarly, the Hurricane Charley article concerns meteorological forecasting accuracy and has no apparent business relevance. Since AG News was constructed through automated web scraping and keyword-based labeling rather than human annotation, a degree of label noise is expected. These errors represent a hard ceiling on achievable accuracy that no model improvement can overcome, as the ground truth itself is incorrect.

Ambiguous samples are articles that could reasonably belong to either category, where even a human annotator might disagree on the correct label. The Amazon/Joyo acquisition is a representative example: it is structurally an M&A story, which would typically fall under Business, yet it concerns an internet retail platform and involves companies whose primary identity is technological. Similarly, the Yahoo pricing article discusses a commercial strategy decision but centers on web hosting and domain registration products. These cases reflect a fundamental limitation of the AG News label schema, which forces a single category onto articles that genuinely straddle two domains. A model trained on this schema will inevitably produce errors on such samples regardless of its representational capacity.

Samples with overlapping keywords are perhaps the most common error type and the most directly attributable to the TF-IDF representation. These are articles whose true category is clear, but whose vocabulary is dominated by terms strongly associated with the other category. Tech company earnings reports are the clearest example: articles about Intuit, Ciena, and Microsoft contain financial language such as "quarterly loss", "forecast", and "government deal" that the model associates with Business, while the entities themselves (software companies, telecom equipment makers) carry strong Sci/Tech signal. The model has no mechanism to weight the framing of an article over the entities it mentions, so whichever signal is stronger in the term frequency wins.

Samples with insufficient context are short or sparsely worded articles where the preprocessed text does not contain enough discriminative signal to make a confident prediction. The Nortel joke column is an extreme case, after preprocessing, almost no meaningful content remains, and the only

signal is a stock ticker. The SUV safety article is similarly sparse after stopword removal, leaving terms like "traffic crash fatality data" and "sport utility vehicle" that do not map cleanly to either Business or Sci/Tech. These errors would likely persist even with a contextual model, since the input itself does not contain sufficient information to determine the correct category.

Finally, some samples are simply difficult to classify and do not fit neatly into any of the above categories. The S2io/Sun Microsystems article, for instance, describes a technology partnership in terms that are equally weighted toward business development and technical infrastructure. The HP holiday shoppers article frames a product strategy announcement in consumer marketing language while discussing flat-screen TVs and media devices. These articles are well-written, correctly labeled, and contain sufficient context, yet the features available to a bag-of-words model do not cleanly distinguish them from the opposite category. Addressing these cases would likely require a model capable of understanding discourse structure and authorial intent, such as a fine-tuned transformer, rather than improvements to the feature representation.

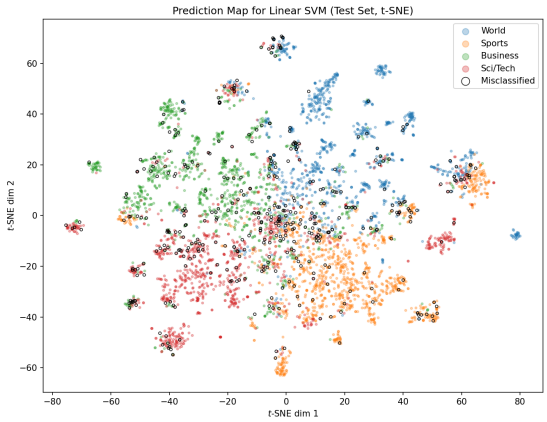

Figure 4: t-SNE Projection of Linear SVM Predictions on Test Set. Correctly classified samples are colored according to their true class, while misclassified samples are outlined in black.

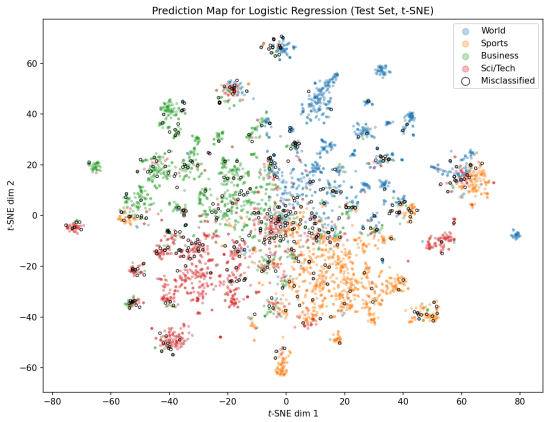

Figure 5: t-SNE Projection of Logistic Regression Predictions on Test Set. Correctly classified samples are colored according to their true class, while misclassified samples are outlined in black.

6 Discussion

It is evident from the results that both models perform quite well on the test set, with accuracies and macro-F1 scores above 0.90. This indicates that the TF-IDF features are effective for this classification task. However, the t-SNE projections of the test set predictions (Figures 4 and 5) reveal there is a lot of overlap between classes. The confusion matrices (Figures 2 and 3) show that there are some misclassifications, particularly between the 'business' and 'sci/tech' classes, which is also reflected in the t-SNE projections where the clusters for these classes overlap more than the others.

After our error analysis, we performed three ablation studies on the dev set to explore the impact of different preprocessing and feature extraction choices on model performance, with full results

reported in Table 4.

The most impactful ablation was stopword removal, which reduced misclassifications by 62 for Logistic Regression (653 vs. 715) and 18 for Linear SVM (646 vs. 664). This is likely because stopwords such as "the", "is", and "of" carry no discriminative signal for topic classification, and their removal allows TF-IDF to assign higher relative weights to informative content words. Notably, keeping stopwords also shifted the optimal SVM hyperparameters from $C = 0.1$ with squared hinge loss to $C = 1.0$ with hinge loss, suggesting the model requires stronger regularisation to cope with the noisier, higher-dimensional feature space introduced by function words.

The n-gram range ablation showed that unigrams alone and unigrams combined with bigrams perform nearly identically. For Logistic Regression, adding bigrams reduced misclassifications by only 22 (673 vs. 651), while for Linear SVM, bigrams actually increased errors marginally (641 vs. 643). This suggests that for this classification task, individual words already provide sufficient discriminative signal, and the additional sparsity introduced by bigrams does not yield meaningful gains.

Finally, lowercasing had a negligible effect on both models, with differences of fewer than 10 misclassifications in either direction. Overall, the modest scale of these differences across all three ablations suggests that while preprocessing choices can influence model performance at the margin, they are not the primary drivers of classification accuracy for this task.

Ablation	**Model**	**Accuracy**	**Macro-F1**	**Misclassified**
With lowercasing	Logistic Regression	0.91	0.91	653
	Linear SVM	**0.92**	0.91	646
Without lowercasing	Logistic Regression	**0.92**	0.91	646
	Linear SVM	**0.92**	**0.92**	643
Unigram only	Logistic Regression	0.91	0.91	673
	Linear SVM	**0.92**	**0.92**	**641**
Unigram + Bigram	Logistic Regression	0.91	0.91	651
	Linear SVM	**0.92**	**0.92**	643
With stopword removal	Logistic Regression	0.91	0.91	653
	Linear SVM	**0.92**	0.91	646
Without stopword removal	Logistic Regression	0.91	0.91	715
	Linear SVM	0.91	0.91	664

Table 4: Ablation study results on the dev set. Misclassification counts are reported alongside accuracy and macro-F1 to highlight differences that aggregate metrics obscure. Bold indicates the best value per column.

We also analyzed the most important features for each class in the Linear SVM model (Figure 6). We found that the feature weights largely make sense: NASA, space, scientist, linux, robot, cybersecurity on the Sci/Tech side. Championship, nba, baseball, olympic, stadium on Sports. Economy, bank, finance, investing, enron, mutual fund on Business. Iraqi, nuclear, arafat, palestinian, darfur on World. The model has genuinely learned meaningful discriminative vocabulary.

It is worth mentioning that we can see clear artifacts from our preprocessing pipeline, such as

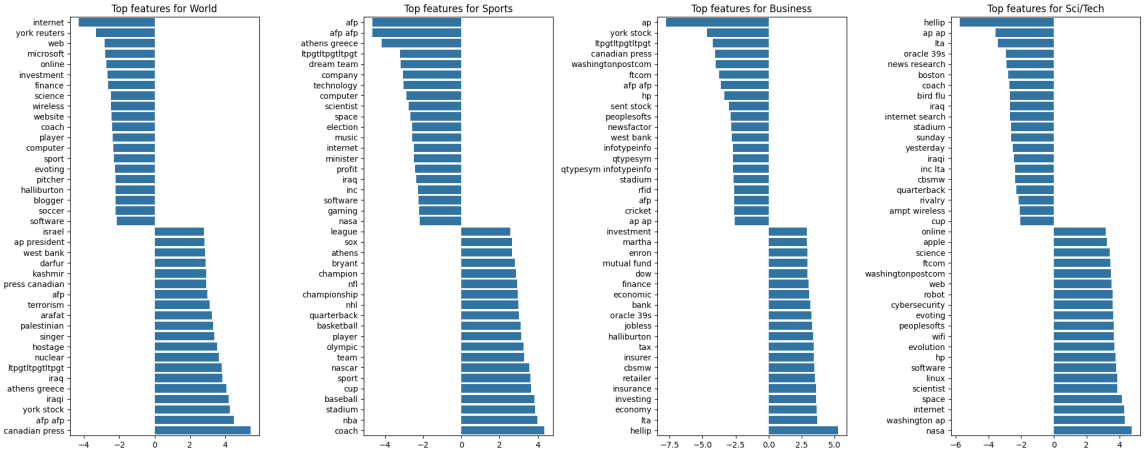

Figure 6: Top 40 discriminative features per class for the Linear SVM model, ranked by coefficient magnitude. Positive values (right) indicate features strongly associated with the given class, while negative values (left) indicate features that push predictions away from that class.

`ltpgtltpgtltpgt`, a residual from HTML paragraph tags (`<p>`) where the entity encoding (`<p>`) was not resolved before punctuation removal, and `oracle 39s`, which originates from the numeric HTML entity `'` for an apostrophe in `Oracle's`. These tokens carry no linguistic meaning and could be eliminated by applying `html.unescape()` prior to preprocessing.

It gets tricky when trying to explain why business and sci/tech are confused. The top features are meaningful for both classes, and there is no overlap in the top features for these classes, so it is not a case that a word like "technology" is pulling an article into both classes. Instead, it must be that some articles have a mix of features from both classes. For example, an article about "nasa" (a top feature for sci/tech) that also mentions "investing" or "economy" (top features for business) might be difficult for the model to classify correctly, especially if the TF-IDF scores for the business-related terms are higher than those for the sci/tech-related terms. This is one of the main limitations of the TF-IDF representation, which does not capture the context in which words are used, and can lead to misclassifications when articles contain a mix of features from different classes.

7 Limitations

Our results point to two interconnected limitations. The first is the bag-of-words assumption underlying TF-IDF. Because the representation weights tokens purely by frequency with no awareness of context, the model cannot distinguish how a word is being used, only whether it appears. This directly explains the overlapping keywords errors we identified: an article containing terms strongly associated with both business and sci/tech will be pulled in both directions, and if the business-related terms happen to carry higher TF-IDF scores, the model will misclassify the article even if its subject matter is primarily technological. The same issue underlies the "simply difficult" category, where the balance of term frequencies failed to reflect the article's overall framing. Contextual encodings such as a fine-tuned transformer would address the two limitaitons; static word embeddings

like Word2Vec would help with dealing with synonyms, but not with nuanced contextual meaning.

The second limitation is the linearity of our classifiers. The t-SNE projections show substantial overlap between the business and sci/tech clusters, which a linear decision boundary cannot cleanly resolve. Non-linear models could in principle learn more expressive boundaries in this region. However, our ablation results suggest that the representation is the primary bottleneck: the gains available from preprocessing changes were modest, which implies that the ceiling imposed by the bag-of-words assumption matters more than the choice of classifier. Switching to a transformer encoder would address both limitations simultaneously, at the cost of the interpretable feature analysis shown in Figure 6.

References

[Zhang et al.(2015)Zhang, Zhao, and LeCun] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems 28 (NIPS 2015)*, 2015.

A Appendix

Table 5: Misclassified samples: *Business* articles predicted as *Sci/Tech*.

Preprocessed Text	**Raw Text (Excerpt)**
teenage rex monster growth tyrannosaurus rex achieved massive size due enormous growth spurt adolescent year	Teenage T. rex's monster growth: Tyrannosaurus rex achieved its massive size due to an enormous growth spurt during its adolescent years.
people eligible get google ipo google billed ipo way everyday people...	Some People Not Eligible to Get in on Google IPO: Google has billed its IPO as a way for everyday people to get in on the process...
rival try turn table charles schwab michael liedtke san francisco ap low price iconoclastic attitude discount stock broker...	Rivals Try to Turn Tables on Charles Schwab: With its low prices and iconoclastic attitude, discount stock broker Charles Schwab Corp. represented an annoying stone in Wall Street's wing-tipped shoes...
dell exit lowend china consumer pc market hong kong reuters dell inc world largest pc maker said monday left lowend consumer pc market china...	Dell Exits Low-End China Consumer PC Market: Dell Inc., the world's largest PC maker, said on Monday it has left the low-end consumer PC market in China...
ibm buy two danish service firm ibm said tuesday acquired pair danish service firm part effort broaden presence scandinavia...	IBM Buys Two Danish Services Firms: IBM said Tuesday it has acquired a pair of Danish IT services firms as part of its effort to broaden its presence in Scandinavia...
microsoft finalises threeyear government deal hot heel 10year strategic partnership london borough newham...	Microsoft finalises three-year government deal: Hot on the heels of its 10-year strategic partnership with the London Borough of Newham...
intuit post wider loss charge reuters intuit inc maker ofthe 1 u tax presentation software turbotax wednesdayposted wider quarterly loss...	Intuit Posts Wider Loss After Charge: Intuit Inc., maker of the No. 1 U.S. tax presentation software TurboTax, posted a wider quarterly loss...
hurricane center 39s projection charley far data show fort lauderdale fla krt despite criticism better anticipated hurricane charley...	Hurricane center's projection on Charley not far off, data show: Despite criticism that it should have better anticipated Hurricane Charley's rapid intensification...
ciena post loss forecast flat sale chicago reuters telecommunication equipment maker ciena corp reported wider loss thefiscal third quarter...	Ciena Posts a Loss, Forecasts Flat Sales: Telecommunications equipment maker Ciena Corp. reported a wider loss for the fiscal third quarter due to slack demand...
report consumer tuning plasma tv firstquarter shipment plasma television united state doubled previous year according research firm isuppli...	Report: Consumers tuning in to plasma TVs: First-quarter shipments of plasma televisions in the United States more than doubled from the previous year...

Table 6: Misclassified samples: *Sci/Tech* articles predicted as *Business*.

Preprocessed Text	**Raw Text (Excerpt)**
intel delay product aimed highdefinition tv san francisco latest series product delay intel corp postponed launch video display chip...	Intel to delay product aimed for high-definition TVs: In the latest of a series of product delays, Intel Corp. has postponed the launch of a video display chip...
yahoo ups ante small business web giant yahoo gambling price cut domain name registration web hosting product make competitive discounter...	Yahoo! Ups Ante for Small Businesses: Web giant Yahoo! is gambling that price cuts on domain name registration and Web hosting products will make it more competitive...
suv live let die new york newly released traffic crash fatality data something everyone debate safety sport utility vehicle	SUVs: Live And Let Die: The newly released traffic crash fatality data have something for everyone in the debate about the safety of sport utility vehicles.
ross store profit plummet 40 percent ap discount retailer ross store inc wednesday said profit fell 40 percent latest quarter due problem new computer system...	Ross Stores Profit Plummets 40 Percent: Discount retailer Ross Stores Inc. said its profit fell about 40 percent in the latest quarter due to problems with a new computer system...
ohio sue best buy alleging used sale ap ohio authority sued best buy co inc thursday alleging electronics retailer engaged unfair deceptive business practice	Ohio Sues Best Buy, Alleging Used Sales: Ohio authorities sued Best Buy Co. Inc. alleging the electronics retailer engaged in unfair and deceptive business practices.
amazon snap china largest web retailer newsfactor amazoncom nasdaq amzn said buy joyocom limited british virgin island company operates largest internet retail web site china...	Amazon Snaps Up China's Largest Web Retailer: Amazon.com has said it will buy Joyo.com Limited, a company that operates the largest Internet retail Web site in China, for $75 million.
deal s2io champing gigabit ottawa local firm say help shrink backup time large data centre growing business thanks alliance sun microsystems inc	Deal has S2io champing at the gigabit: A local firm that says it can help shrink backup times at large data centres is growing its business thanks to an alliance with Sun Microsystems Inc.
mom 2005 released manufacturing microsoft wednesday announced release manufacturing microsoft operation manager mom 2005 workgroup edition...	MOM 2005 Released to Manufacturing: Microsoft announced the release to manufacturing of Microsoft Operations Manager (MOM) 2005 and MOM 2005 Workgroup Edition...
hp tempt holiday shopper sight sound computerhardware giant best known product pc printer laid plan become brandname consumer electronics product...	HP to tempt holiday shoppers with sights and sounds: The computer-hardware giant laid out its plan to become a brand-name in consumer electronics products such as flat-screen TVs and music players...
canadian invasion week ago story nortel nyse nt asked people submit canadian joke good place reveal winner	A Canadian Invasion: A few weeks ago, in a story on Nortel (NYSE: NT), I asked people to submit a Canadian joke to me.